

Benchmarking motor imagery algorithms for pediatric users of brain-computer interfaces

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Abstract—Brain-computer interfaces (BCIs) can enable opportunities for self-expression and life participation for children with severe neurological disabilities. Unfortunately, the development and evaluation of state-of-the-art algorithms has largely neglected pediatric users. This work tests 12 state-of-the-art algorithms for motor imagery classification on three datasets of typically developing pediatric users ($n=94$ ages 5-17). When all datasets were combined, there were no significant differences between most non-deep learning algorithms, with all having a mean AUC score of 0.64 or 0.65. All the non-deep learning algorithms significantly outperformed the deep learning algorithms, which can be partially attributed to a lack of hyperparameter tuning. The best of the deep learning algorithms was ShallowConvNet, with a mean AUC score of 0.57. Of the algorithms tested, only the filter bank common spatial pattern (FBCSP) and ShallowConvNet exhibited significant age effects. This general lack of age effects, combined with examples of children as young as 6 having AUC scores as high as 0.8, provides evidence that young children are capable of producing measurable motor imagery activations. The age effects that were present for some algorithms suggest that the changing EEG patterns associated with development could have a measurable impact on classification algorithm outcomes, and such algorithms should be evaluated to ensure that they are not performing disproportionately poorly for younger children. This work serves as a first step towards ensuring that the state-of-the-art improvements in BCI classification can be evaluated, and where necessary, adapted to meet the needs of pediatric users.

I. INTRODUCTION

Brain-computer interfaces (BCIs) hold the potential for users with disabilities to enhance their opportunities for independence and life participation [1]. This includes children with severe neurological disabilities who report high satisfaction from setting and achieving their own goals with the use of a BCI [2]. We know children are capable of using BCIs, including those based on motor imagery (MI) [3]. However, pediatric populations are often excluded from the development and evaluation of BCIs [4], creating opportunities for bias in what is truly state-of-the-art.

BCIs are multifaceted systems which include signal collection and processing, feature extraction, classification, and feedback. All parts of this system need to be designed and evaluated to ensure that they work for pediatric populations [5]. Electroencephalography (EEG) based MI BCI has been tested for children with Cerebral Palsy (CP) [6], [7] and typically developing children [3], [8], [9], [10]. Of these 6 studies, three are based on proprietary EMOTIV software (EMOTIV Inc., San Francisco, USA) [6], [3], [7], one is based

on the common spatial pattern (CSP) [8], and two are based on Riemannian Geometry (RG) [9], [10]. In these studies, there is little justification for their chosen algorithm, except to reference its performance in adult BCI studies, with the implied assumption that the algorithm will also perform well for children. It is important to test this assumption, knowing that children's EEG varies significantly with development in measures of frequency, topography, and power [11].

Differences in MI BCI task performance based on age have been reported in work by Zhang et al. and Kelly et al. [3], [10]. These differences were attributed to evidence that children are developing their abilities to perform motor imagery between the ages of 7-12 [12]. However, there have been no significant correlations between BCI calibration accuracy in the three recent MI BCI studies with children [8], [9], [10]. This suggests that children are able to execute measurable MI intent, but possibly without the consistency required for functional use.

The purpose of this work is to evaluate 12 state-of-the-art algorithms for use with pediatric MI BCI so that future studies can make more informed choices about which algorithm to use. In addition to evaluating the performance of different algorithms, we will investigate potential age effects associated with different algorithms to determine which algorithms may be most susceptible to development-related changes in EEG.

II. METHODS

A. Datasets

This work includes 3 left-right MI datasets from 2 prior BCI studies, the details for which can be found in Table I [9], [10]. The key difference between them is in their montages and their experimental design. FatigueMI uses a whole-head montage [9]. OptStdMI and OptGameMI use a more compact montage concentrated over the motor cortex [10]. FatigueMI and OptStdMI were collected using a standard MI paradigm, where the intended target (left or right hand) was indicated by a square enlarging on the intended side, and feedback on the BCIs classification was delivered by increasing the colour saturation of the square [9], [10]. OptStdMI indicated the intended target by placing bananas to the left or right of a monkey, feedback was delivered by the monkey moving to the left or right, collecting the bananas. OptStdMI and OptGameMI share 30 participants [10]. The order in which these participants completed OptStdMI and OptGameMI was randomized to negate learning effects [10].

TABLE I
THREE PEDIATRIC DATASETS USED FOR ALGORITHM EVALUATION

Dataset	# Participants	Age Range (Mean)	# Female	# Channels	Channel Labels	Epoch Length (s)	# Epochs	Ref.
FatigueMI	32	7-16 (10)	20	20	A2, C3, C4, Cz, F3, F4, F7, F8, Fp1, Fp2, Fz, O1, O2, P3, P4, Pz, T3, T4, T5, T6	2	108	[9]
OptStdMI	31	5-17 (11)	13	16	C1, C2, C3, C4, C5, C6, CP1, CP2, CP3, CP4, CPz, Cz, FC3, FC4, FCz, Pz	1.5	120	[10]
OptGameMI	31	5-17 (11)	14	16	C1, C2, C3, C4, C5, C6, CP1, CP2, CP3, CP4, CPz, Cz, FC3, FC4, FCz, Pz	1.5	120	[10]

TABLE II
ALGORITHMS TO BE EVALUATED.

Algorithm	Description	Ref.
MDM	Riemannian Minimum Distance to Mean	[13]
Tangent Space LDA	Tangent space projection of covariance matrix with shrinkage linear discriminant analysis (LDA)	[13]
Tangent Space SVM	Tangent space projection of covariance matrix with support vector machine (SVM)	[13]
CSP + LDA	CSP supervised spatial filter with shrinkage LDA	[14]
CSP + SVM	CSP supervised spatial filter with SVM	[14]
FBCSP + LDA	Filter-bank CSP with LDA	[15]
FBCSP + SVM	Filter-bank CSP with SVM	[15]
EEGITNet	Convolutional Neural Network (CNN) with EEGITNet architecture, downsampled to 128 Hz	[16]
EEGNeX	CNN with EEGNeX architecture, downsampled to 128 Hz	[17]
EEGNet_8_2	CNN with EEGNet architecture, downsampled to 128 Hz	[18]
EEGTCNet	CNN with EEGTCNet architecture, downsampled to 250 Hz	[19]
ShallowConvNet	CNN, downsampled to 250 Hz	[20]

B. Algorithms

A representative, but non-exhaustive, subset of algorithms was selected from a recent study evaluating MI algorithms with MOABB [21]. This subset was refined by excluding algorithms which rely on data augmentation, transfer learning, or adaptive learning. This was decision was intended to limit the scope of this work while maintaining a representative sample of common spatial pattern (CSP), Riemannian Geometry (RG), and Deep Learning (DL) algorithms. Details about each implementation can be found in Table II. Because the data includes children across a range of development, and EEG changes with development, it is unlikely that hyperparameter tuning on the dataset level would generalize well for each individual. Alternatively, hyperparameter tuning on the individual level could be too time-consuming for real-world BCI use. For this reason, all algorithms were assessed for their performance as an out-of-the-box solution, without optimization.

Each algorithm was subjected to identical pre-processing and evaluation to ensure consistency. Pre-processing consisted of band-pass filtering with a pass-band of 8-32 Hz. Filter-bank

TABLE III
MEAN ALGORITHM AUC SCORES (STANDARD DEVIATION)

Pipeline	FatigueMI	OptGameMI	OptStdMI	Combined
MDM	0.64 (0.12)	0.65 (0.16)	0.64 (0.14)	0.64 (0.14)
Tangent Space LDA	0.65 (0.13)	0.65 (0.18)	0.64 (0.14)	0.65 (0.15)
Tangent Space SVM	0.67 (0.13)	0.64 (0.18)	0.63 (0.14)	0.64 (0.15)
CSP + LDA	0.65 (0.13)	0.65 (0.16)	0.64 (0.14)	0.65 (0.14)
CSP + SVM	0.66 (0.13)	0.66 (0.16)	0.64 (0.15)	0.65 (0.15)
FBCSP + LDA	0.66 (0.13)	0.68 (0.13)	0.64 (0.12)	0.65 (0.13)
FBCSP + SVM	0.66 (0.13)	0.66 (0.14)	0.62 (0.12)	0.64 (0.13)
EEGITNet	0.51 (0.08)	0.54 (0.12)	0.55 (0.09)	0.53 (0.10)
EEGNeX	0.52 (0.06)	0.52 (0.11)	0.51 (0.09)	0.51 (0.09)
EEGNet_8_2	0.54 (0.09)	0.60 (0.15)	0.57 (0.13)	0.57 (0.13)
EEGTCNet	0.50 (0.07)	0.54 (0.12)	0.54 (0.10)	0.53 (0.10)
ShallowConvNet	0.58 (0.11)	0.62 (0.15)	0.61 (0.12)	0.61 (0.12)

based methods used 6 non-overlapping frequency windows 8-12 Hz, 12-16 Hz, 16-20 Hz, 20-24 Hz, 24-28 Hz, and 28-32 Hz. Deep learning methods were downsampled to either 128 Hz or 250 Hz to match their MOABB implementations. For consistency, hyperparameters were taken from a recent study which evaluated multiple pipelines using the MOABB framework [21]. DL models used a sparse categorical cross-entropy loss-function, Adam optimization (learning rate = 0.0009), 300 epochs, and a batch size of 64. Early stopping (patience = 75) and learning rate reduction on plateau (factor = 0.5, patience = 75) were used for all DL models. All algorithms were evaluated using 5-fold cross validation and scored on the area under the curve (AUC) of their receiver operating characteristic (ROC). All evaluations were completed using MOABB [22].

III. RESULTS

The mean AUC scores and standard deviations from the evaluation of each dataset and each pipeline are shown in Table III. The combined column shows the AUC scores for all individuals from all datasets combined. Mean AUC scores are tightly distributed around 0.65 for all non-DL algorithms. Of the DL algorithms, ShallowConvNet performs the best, with a combined mean AUC score of 0.61. Other DL algorithms range from 0.51 (EEGNeX) to 0.57 (EEGNet_8_2). Standard deviation for all algorithms is high, between 0.1 and 0.15. Scores are similar across datasets for the same pipeline. Mean

training duration for all non-algorithms was less than 1 s, mean training duration for DL algorithms ranged from 17-46 s.

To compare between algorithms, Fig. 1 shows the significant differences between algorithms on the combined data using the Wilcoxon Signed Rank test with false discovery rate correction. It shows there are no significant differences between most of the non-DL algorithms, except for CSP and FBCSP, which seem to work better with SVM and LDA, respectively. Fig. 1 shows significant differences, indicating that all non-DL algorithms outperformed all DL algorithms in this evaluation. Finally, this figure shows that ShallowConvNet significantly better than all other DL algorithms, with EEGNet_8_2 performing second best.

To evaluate potential age effects, the AUC scores across all datasets were correlated with participant age to determine if there were any age effects for the different algorithms. The scatter plot of AUC v. age for the MDM classification in Fig. 2A shows that performance is widely distributed, with low and high-scores at all ages. For example, there are multiple individuals under 8 years of who achieved AUC scores of 0.8 or above, matching the performance of high scoring adult users. Fig. 2B also shows significant age effects were found for FBCSP + LDA, FBCSP + SVM, and ShallowConvNet, with older children performing better than their younger counterparts.

IV. DISCUSSION

There were few significant differences between the AUC scores of non-DL pipelines, which is reassuring for pediatric BCI studies which have already been conducted using either RG or CSP based algorithms. The most prominent difference is between the non-DL algorithms and the DL algorithms. This can be partially attributed to not tuning the DL hyperparameters for each individual. However, the training duration of the DL methods shows why hyperparameter optimization may not be possible for each individual, with DL training taking 17-46 s for each individual, optimization of multiple hyperparameters could take longer than what is feasible within a real-world BCI session. This should not be seen as a critique of DL methods for pediatric BCI, but simply a warning that these algorithms should not be expected to work out-of-the-box, compared to non-DL methods. Future work should evaluate how hyperparameters generalize across pediatric datasets to determine if individual hyperparameter tuning is required, and assess how individual training times can be reduced with transfer learning. Of the DL models, ShallowConvNet had the best out-of-the-box performance on this sample of data, suggesting it is a feasible DL solution for pediatric MI BCI. A limitation of these performance measurements is that they neglect the learning rate of the different algorithms. This is particularly relevant to pediatric BCI, where shorter calibration sessions are especially desirable, as children can easily become bored or distracted by the calibration. Investigating the learning rate could provide a differentiating factor for deciding between algorithms and should be investigated in the future.

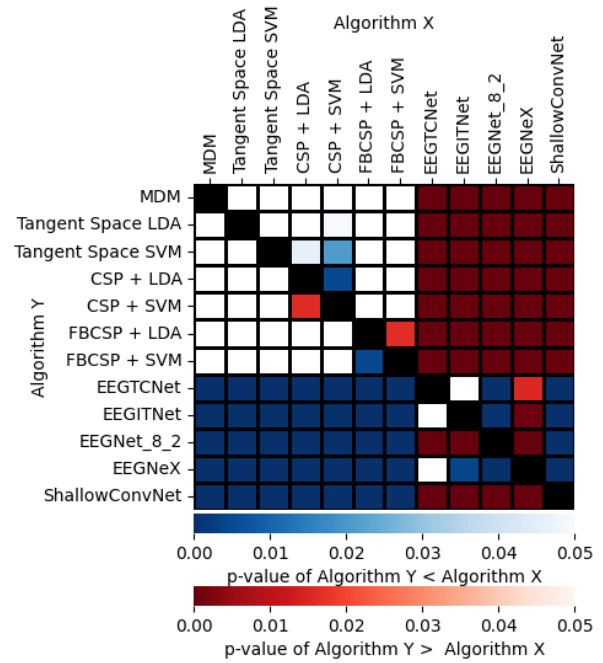


Fig. 1. Heatmap of significant differences in AUC scores for the combined dataset. p-values are based on two one-sample Wilcoxon Signed Rank tests, one to determine if Y is greater than X and one to determine if X is greater than Y. p-values were corrected with false discovery rate (FDR) correction.

Our finding that most of the algorithms, excluding FBCSP algorithms and ShallowConvNet, do not exhibit age effects is supportive of prior evidence [8], [9], [10]. This suggests that children as young as 6 are capable of repeatedly producing a measurable MI action, at least for short periods of time. Age effects seen in longer, task-based, MI BCI trials could be a result of young children being less able to sustain, rather than initiate, the correct intention. In this offline analysis, the presence of age effects with the FBCSP algorithms could be a result of varying EEG bandpower frequencies through development and should be explored further. Additional investigations could include determining if the width, overlap, or number of bands influences the performance score of the FBCSP algorithms for children. This future work should optimize frequency band selection to ensure that filter bank methods help, rather than hinder, young children. The age effect seen with the ShallowConvNet should also be explored in greater detail, with an initial point exploring if the apparent age effect could be reduced with hyperparameter tuning.

This paper shows how well different algorithms perform on pediatric data, but it should not be seen as exhaustive. In addition to the aforementioned hyperparameter tuning and optimization of filter banks, there are many other state-of-the-art techniques which could improve classification for pediatric BCI. These include data augmentation, feature engineering, and transfer learning, all of which have been studied for adult users, but have yet to be tested for pediatric users. Further work should be done to validate state-of-the-art BCI methods for pediatric use.

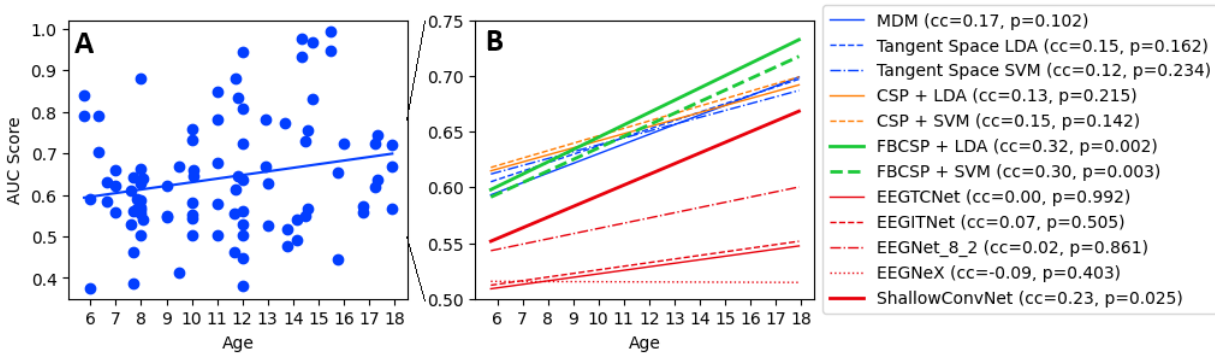


Fig. 2. Age effects of different algorithms. A) A scatter plot of AUC score v. age for the MDM algorithm with a line of best fit. B) An exploded view showing the lines of best fit for all algorithms. All RG algorithms are blue, all CSP algorithms are orange, all FBCSP algorithms are green, and all DL algorithms are red. Significance is indicated by greater line thickness. Correlation coefficients (cc) and p-values of the Spearman correlation are included in the legend.

V. CONCLUSION

We evaluated 12 commonly used algorithms for MI BCI and evaluated them retrospectively across 3 datasets of typically developing children. All algorithms received common pre-processing and evaluation to ensure fairness across algorithms. Of the tested algorithms, all the non-DL algorithms performed similarly in terms of AUC score, and all performed better than the DL algorithms. Of the DL algorithms, Shallow-ConvNet performed the best, but also exhibited a significant age effect, performing disproportionately poorly for younger children. FBCSP algorithms also had significant age effects, which could be related to the way in which the filter banks were designed and should be further investigated. This work represents a first step in evaluating state-of-the-art methods, which have only been tested on adults, for pediatric use.

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